

IIT BOMBAY
CHEMICAL ENGINEERING

Growth Stoichiometry

The Black-Box Model, Substrate Partitioning & Elemental Balances

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Why Growth Stoichiometry Matters

Every fermenter design question comes back to one thing: how much of everything is needed to make how much of what.

How much substrate and O₂ must be fed

To reach a target cell density, and by when?

How much heat must be removed

Growth is exothermic — no balance, no cooling design.

How much product forms, at what cost

How many moles of substrate does one mole of product actually take?

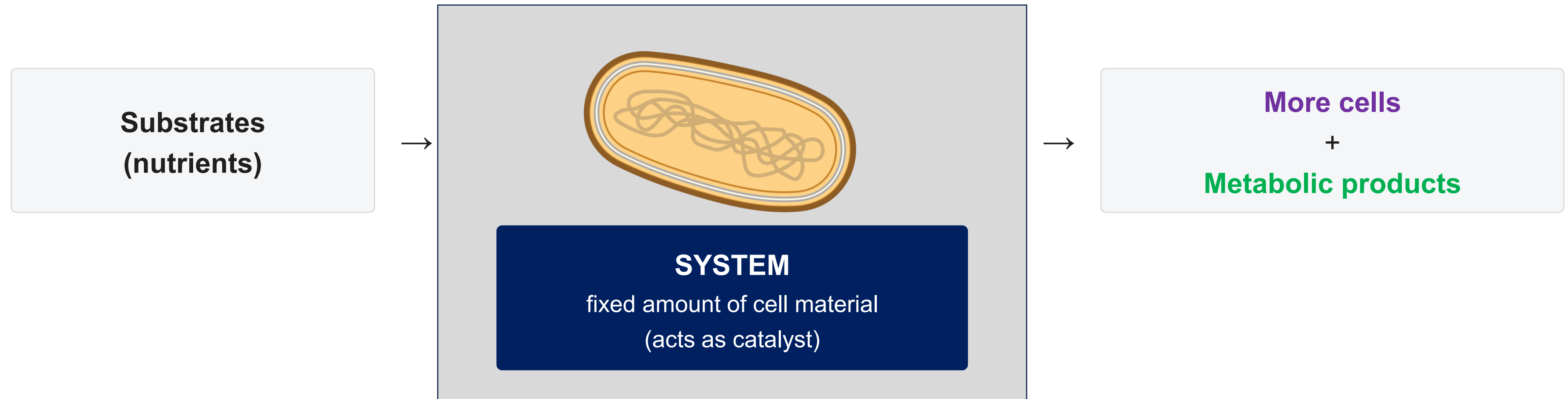
How is the growth medium designed

so that no single nutrient runs out before the others?

Every one of these questions needs a balanced equation for growth, it's the same idea as balancing any chemical reaction. The difference: here, the “product” (biomass) has a composition that must be measured experimentally, not looked up in a table. That's exactly what the rest of this deck builds, step by step.

Zooming Out: Cell Growth as a Black Box

For stoichiometry, cell material itself can be treated as the catalyst of its own growth.



This “black box” view underlies stoichiometric medium design: to grow a target mass of cells (X) with elemental weight fraction w_i , at minimum $w_i(X)$ of element i must be supplied in the nutrients.

From Black Box to Numbers: Yield Coefficients

The simplest way to quantify the black box: ratios of what goes in to what comes out.

$$Y_{X/S} = \Delta X / \Delta S$$

cell mass produced per unit substrate consumed (g/g or Cmol/Cmol)

$$Y_{P/S} = \Delta P / \Delta S$$

product formed per unit substrate consumed

$$Y_{X/O_2} = \Delta X / \Delta O_2$$

cell mass produced per unit O₂ consumed (aerobic growth only)

$$RQ = \Delta CO_2 / \Delta O_2$$

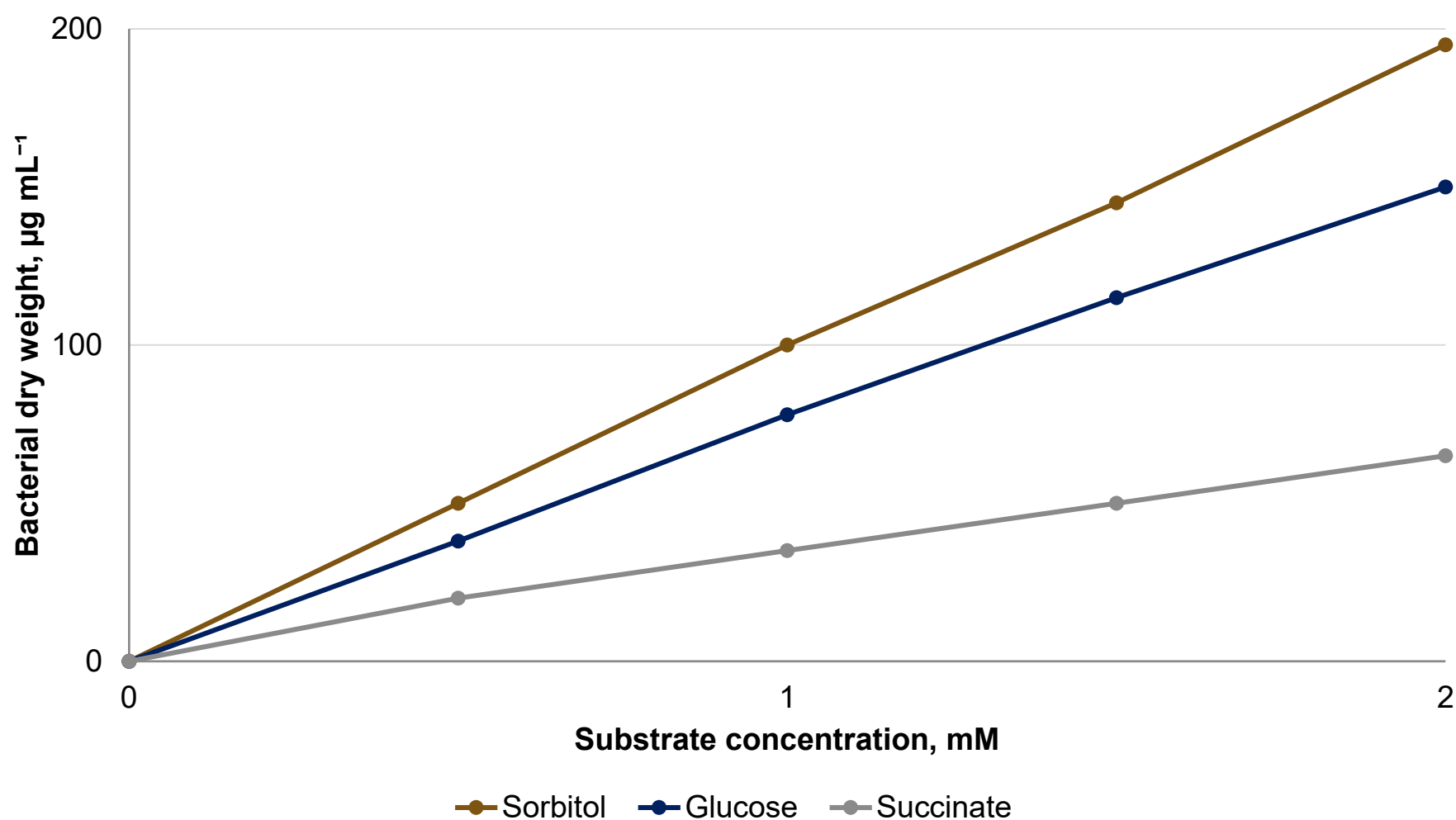
respiratory quotient — easily measured online from off-gas analysis

These are the black box's measurable outputs. Sections 5.10.2–5.10.3 explain them mechanistically, atom by atom.

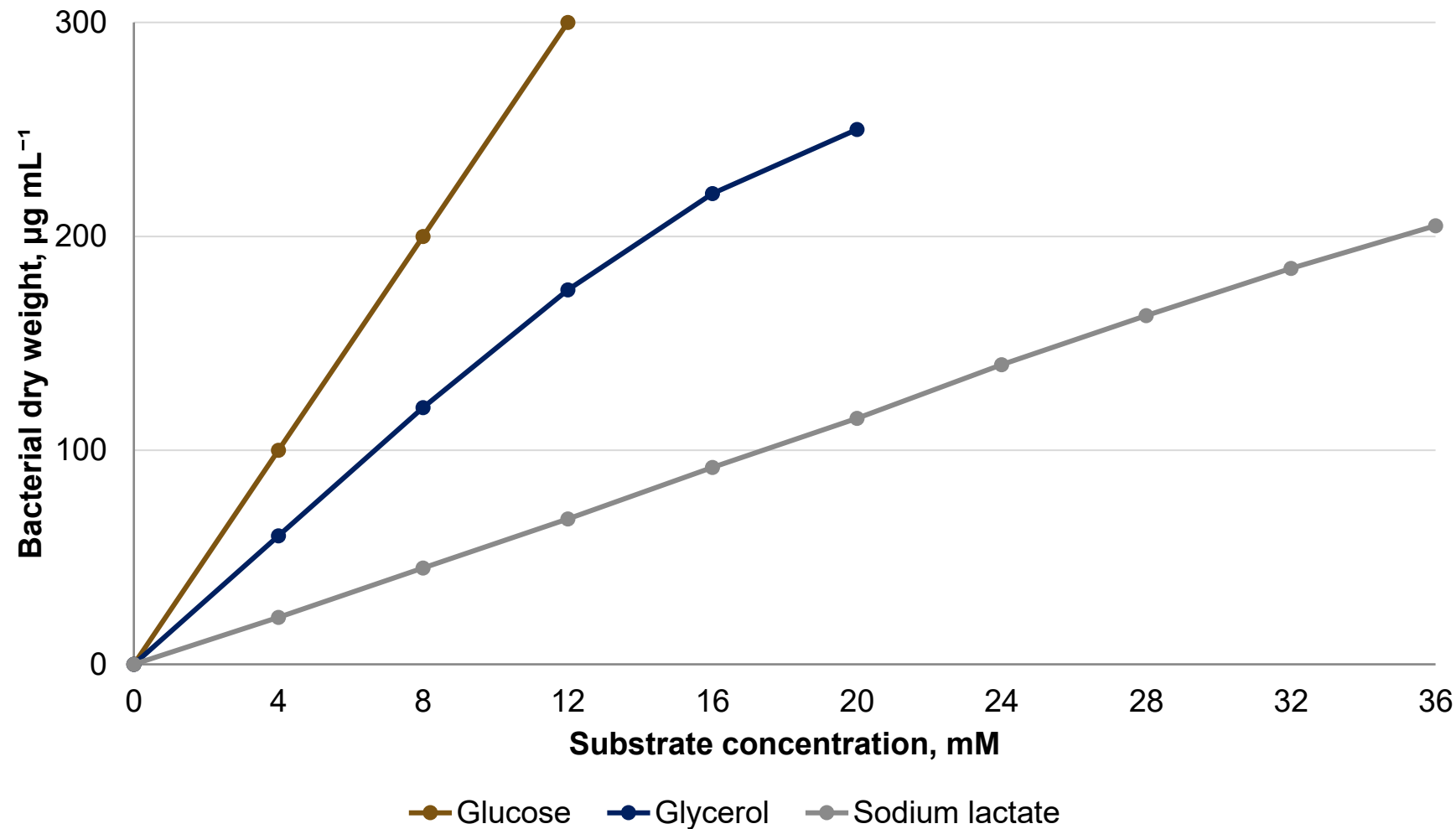
Growth Yields on Different Carbon Sources (Fig. 5.29)

Each line's slope is an experimentally measured $Y_{x/s}$ substrate identity changes the yield.

(a) Aerobic — *Aerobacter aerogenes*



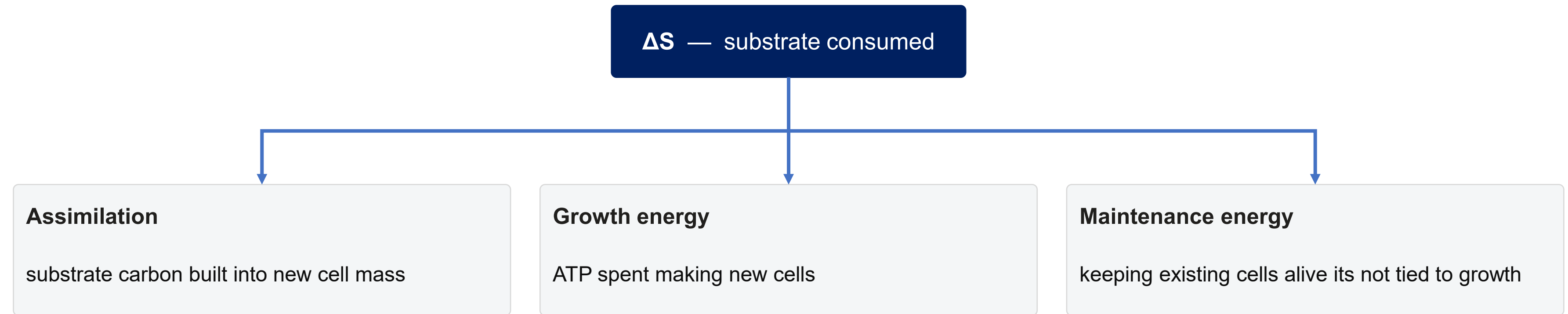
(b) Anaerobic — *Propionibacterium pentosaceum*



Slopes are apparent $Y_{x/s}$: sorbitol > glucose > succinate (aerobic); glucose > glycerol > lactate (anaerobic) that is more reduced substrates tend to support higher yields.

Where Does the Substrate Actually Go?

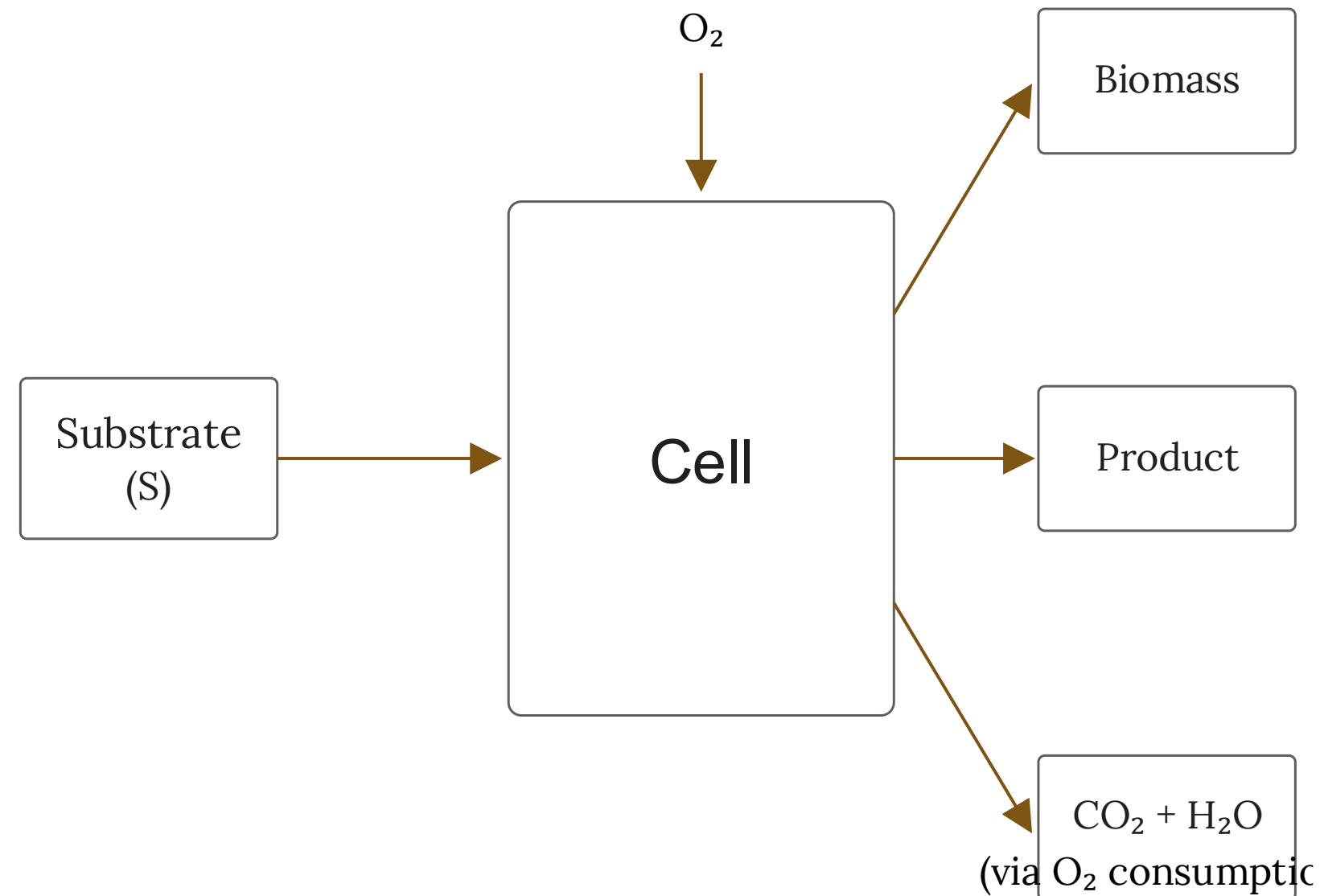
Not all consumed substrate is used to build new cells; some is used for energy and maintenance.



$$(\Delta S) = (\Delta S)_{\text{assimilation}} + (\Delta S)_{\text{growth energy}} + (\Delta S)_{\text{maintenance energy}} \quad (\text{Eq. 5.47})$$

$$1/Y_{X/S} = (\Delta S)_{\text{assim}}/(\Delta X) + (\Delta S)_{\text{growth}}/(\Delta X) + (\Delta S)_{\text{maint}}/(\Delta X) \quad (\text{Eq. 5.48})$$

Motivation: Tracking Electrons in Bioreactions



- Mass balances (C, H, O, N) don't directly show energy/electron routing
- Aerobic respiration uses O₂ as terminal electron acceptor
- Need a way to check if measured yields (X, P, O₂) are consistent

Degree of Reduction (γ): Definition

PLAIN-LANGUAGE DEFINITION

γ = electrons available per C-mole, relative to CO_2 , H_2O , NH_3 as zero-reference and $\text{H} = 1$.

FORMULA

For $\text{CH}_a\text{O}_b\text{N}_c$:

$$\gamma = 4 + a - 2b - 3c$$

C: +4 H: +1 O: -2 N: -3

WORKED EXAMPLE – GLUCOSE (CH_2O)

$$a = 2, b = 1, c = 0$$

$$\gamma = 4(1) + 1(2) - 2(1) - 3(0)$$

$$\gamma = 4 + 2 - 2 - 0$$

$$\gamma = 4.00$$

Degree of Reduction of Key Compounds

COMPOUND	FORMULA	Γ
Glucose	CH ₂ O	4.00
Ethanol	CH ₃ O	6.00
Glycerol	CH _{2.67} O	4.67
Lactic acid	CH ₂ O	4.00
Biomass, typical	CH _{1.8} O _{0.5} N _{0.2}	approx. 4.20
O ₂	O ₂ (per mol)	-4.00
CO ₂ / H ₂ O / NH ₃	reference	0

O₂ has negative γ because it's the electron acceptor, not donor.

Elemental Composition & Degree of Reduction (Table 5.11)

Representative literature values for common organisms — with a CHO cell row added for mammalian bioprocessing.

Organism	Empirical formula (per C-mol)	γ (per C-mol)
Escherichia coli	$\text{CH}_{1.77}\text{O}_{0.49}\text{N}_{0.24}$	4.07
Saccharomyces cerevisiae	$\text{CH}_{1.64}\text{O}_{0.52}\text{N}_{0.16}$	4.12
Candida utilis	$\text{CH}_{1.83}\text{O}_{0.54}\text{N}_{0.17}$	4.24
Aerobacter aerogenes	$\text{CH}_{1.75}\text{O}_{0.43}\text{N}_{0.22}$	4.23
“Average” biomass (Roels)	$\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$	4.20
CHO cells (mammalian) *	$\text{CH}_{1.75}\text{O}_{0.55}\text{N}_{0.20}$	4.05

$\gamma = 4C + H - 2O - 3N$ per C-mol (“available electrons” relative to CO_2 , H_2O , NH_3). * CHO row is a representative literature estimate compiled for this deck (typical mammalian-cell composition) — not a verified reproduction of the textbook's exact entry.

Why Degree of Reduction? γ as an Electron Currency

γ counts available electrons per C-mole — one currency for balancing any redox-neutral growth or fermentation equation.

$$\gamma = 4C + H - 2O - 3N \quad (\text{per mole, relative to CO}_2, \text{H}_2\text{O, NH}_3 \text{ references})$$

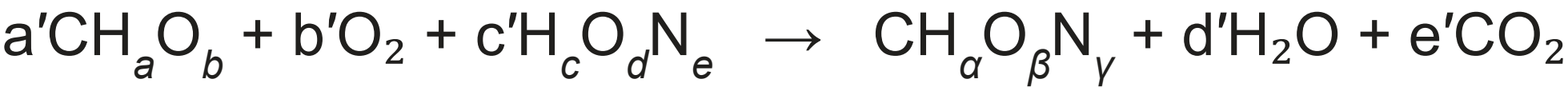
Typical values: glucose $\gamma = 4.0$, ethanol $\gamma = 6.0$, biomass $\gamma \approx 4.0\text{--}4.2$ (table above), CO_2 $\gamma = 0$, O_2 accepts 4 electrons per mole.

$$\gamma_{\text{substrate}} \times (\text{mol substrate}) = \gamma_{\text{biomass}} \times (\text{mol biomass}) + \gamma_{\text{product}} \times (\text{mol product}) + 4 \times (\text{mol O}_2)$$

electrons in = electrons in biomass + electrons in product + electrons handed to O_2 (aerobic) or to reduced by-products (anaerobic)

Elemental Material Balances for Growth

Balancing every C, H, O, and N atom between substrate, nitrogen source, and biomass (growth only).

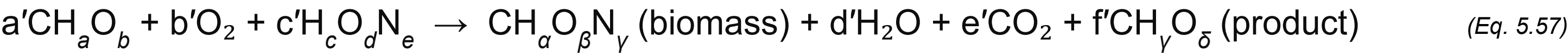


Symbol	Meaning
$a'-e'$	stoichiometric coefficients — fixed by balancing C, H, O, N and available electrons
CH_aO_b	substrate, per mole of carbon (e.g. glucose as CH_2O)
$\text{H}_c\text{O}_d\text{N}_e$	nitrogen source (e.g. NH_3 , NO_3^-)
$\text{CH}_\alpha\text{O}_\beta\text{N}_\gamma$	empirical formula of dry cell mass (biomass), per mole of carbon

4 elemental balances (C, H, O, N) + 1 degree-of-reduction (electron) balance → 5 equations for the 5 unknowns (a' , b' , c' , d' , e'), once the biomass composition (α , β , γ) and substrate / N-source formulas are known.

Product Formation Stoichiometry

Adding an excreted product reopens a degree of freedom that is fixed by the measured product yield.



Symbol	Meaning
$a'-f'$	stoichiometric coefficients — fixed by balancing C, H, O, N and available electrons
CH_aO_b	substrate, per mole of carbon (e.g. glucose as CH_2O)
$\text{H}_c\text{O}_d\text{N}_e$	nitrogen source (e.g. NH_3 , NO_3^-)
$\text{CH}_\alpha\text{O}_\beta\text{N}_\gamma$	empirical formula of dry cell mass (biomass), per mole of carbon
$\text{CH}_\gamma\text{O}_\delta$	empirical formula of the excreted product, per mole of carbon

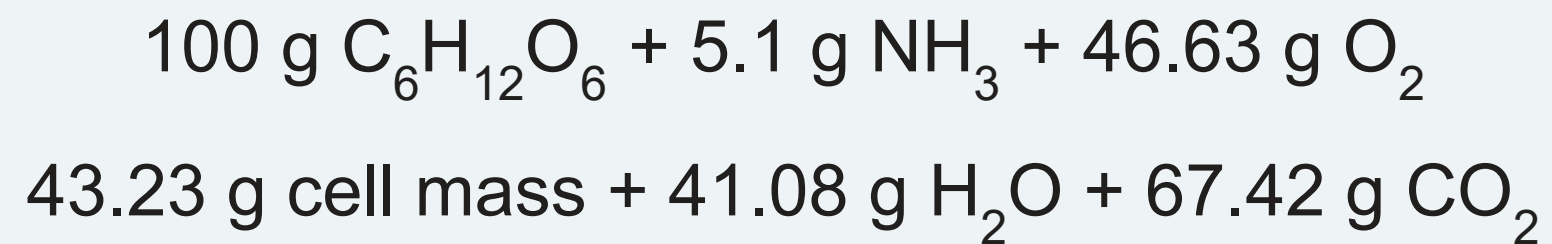
6 unknowns ($a'\dots f'$) now need the 4 elemental balances + 1 electron balance + one measured yield (e.g. the product yield f' , or the product yield coefficient $Y_{p/s}$) — the same missing-degree-of-freedom logic as Problem 2.7(a), one level up.

Writing every formula per mole of carbon keeps the balance dimensionally consistent regardless of molecule size.

Stoichiometry: Aerobic vs. Anaerobic Species

To conveniently calculate nutrient consumption, aeration, and heat-transfer rates, a general stoichiometric equation can be formulated by assigning the cell an apparent molecular formula.

(a) Calculate the cell formula for growth of yeast on sugar, given the following balance (1 mol of cell produced):



(b) If the above reaction produces 0.30017 “mol” of cell, what cell formula results?

Note: *elemental (C, H, O, N) and mass balances on the part-(a) reaction give the mole quantities of each species; dividing by 0.30017 mol cell normalizes the empirical formula $C_aH^bO^cN^d$ to one mole of cell.*

(c) Anaerobic fermentations also yield partially oxygenated by-products. Keeping the cell formula from (b), find the unknown coefficients (mole quantities) below:

0.55556 glucose + α ammonia

→ β cell mass + 0.05697 glycerol

+ γ butanol (alcoholic product)

+ ε succinic acid (acid product)

+ 0.01164 water + 1.03076 carbon dioxide

+ 1.00380 ethanol

Coefficients α , β , γ , and ε are the unknowns to be solved by elemental balance

Cheat Sheet — Growth Stoichiometry

Every formula from this deck, one page.

Black box & yield coefficients

$$Y_{X/S} = \Delta X / \Delta S \quad Y_{P/S} = \Delta P / \Delta S$$

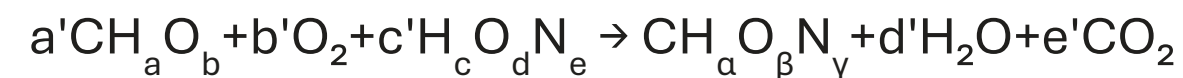
$$Y_{X/O_2} = \Delta X / \Delta O_2 \quad RQ = \Delta CO_2 / \Delta O_2$$

Substrate partitioning (Eqs. 5.47–5.48)

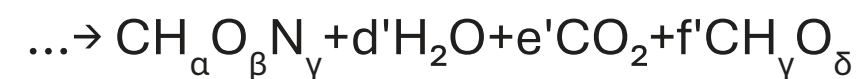
$$\Delta S = \Delta S_{\text{assim}} + \Delta S_{\text{growth}} + \Delta S_{\text{maint}}$$

$$1/Y_{X/S} = \Delta S_{\text{assim}} / \Delta X + \Delta S_{\text{growth}} / \Delta X + \Delta S_{\text{maint}} / \Delta X$$

Elemental balance — growth only (5.10.2)



Elemental balance — with product (5.10.3, Eq. 5.57)



Degree of reduction

$$\gamma = 4C + H - 2O - 3N \quad (\text{per mole})$$

$$\gamma_{\text{sub}} \times n_{\text{sub}} = \gamma_{\text{cell}} \times n_{\text{cell}} + \gamma_{\text{prod}} \times n_{\text{prod}} + 4 \times n_{O_2}$$

Typical γ values (per C-mol):

Glucose (CH₂O): $\gamma = 4.0$

Ethanol (CH₃O_{0.5}): $\gamma = 6.0$

CO₂: $\gamma = 0$

Biomass: $\gamma \approx 4.0\text{--}4.2$

O₂: accepts 4 e[−] / mol